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Trailblazing and Cost Effective Approach for Activation of Inactive Oil Wells Leveraging Mindset of 100x and Single Barrel Hydrocarbon Counts

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Abstract

This paper introduces an innovative practice and improvised initiative developed by Company Asset to reactivate inactive wells that are unable to flow with lift gas. The solution involved fixing bullhead/rocking lines for gas lift wells.

In accordance with operational conditions and production mandates, the necessity of well activation was re-evaluated and successfully executed through this innovative approach. This solution significantly reduced the costs associated with well intervention and ensured the well's readiness without causing production deferment.

The commissioning of gas lift wells presents challenges in unloading the liquid column to allow lift gas from the annulus into the tubing. Occasionally, this process is unsuccessful, leading to back pressure on the gas side and preventing lift gas flow into the well. To address this issue, a 1-inch piping connection was established between the gas side and the oil side. Utilizing the available piping for rocking, rather than relying on conventional external sources, resulted in significant time and cost savings as well as timely activation of the well to enhance production. The wells have been successfully rocked.

The behavior of gas lift wells was analyzed by leveraging the advantage of bullheading line availability at the wellhead, which connects the gas line with the oil production line and includes manual isolation valves. Integrity checks and the sizing of the valve and line were conducted, providing confidence for utilization.

The following steps were safely executed:

- Commencing risk assessment for utilizing existing piping
- Evaluating gas lift wellheads' process safety protections
- Developing safe operating procedures, scenarios, and schematics

Lessons learned highlighted the benefits of minimizing external resources and reducing time consumption in resuming production. The innovative solution of utilizing the bullheading/rocking line proved to be cost-effective. The benefits achieved from this exercise, which will continue to be advantageous in the future, include:

- Reduction of greenhouse gas emissions (3 Metric Tons CO2 equivalent)
- Avoidance of intervention costs amounting to 1 million USD
- Production gain of 10,000 barrels of oil, preventing production loss from gas lift wells
- Well availability exceeding 98%
- Maintenance of inactive strings KPI below 2%
- Ensuring well integrity
- Achievement of project milestones on time
- Avoidance of costs and time incurred due to prolonged downtime.

The implemented solution is highly effective in ensuring continuous production, reducing GHG emissions leading to NET zero mindset and can be replicated to other existing facilities in the company and worldwide. Adding simple piping at the wellhead resulted in activating the wells without using external resources.

This implementation will enhance the availability of wells in a short time span and proved helpful in contributing to the pillar of Sustainability – Reduction in flaring/GHG emissions.

Introduction

Gas lift is one of several processes used to artificially lift oil or water from wells where there is insufficient reservoir pressure to produce the well.

The lift gas is typically provided from a compressor station. The gas lift let-down station is supplied with compressed gas where the pressure is let down from 5,800 to 2,610 psig and fed into a dedicated buried network to the gas lift oil producer wells completed in various parts of the field.

The Gas lift main header and the sub headers supplying various gas lift wells also have provisions to hook up temporary pigging facilities to carry out internal inspection.

The gas is injected into the well at a controlled pressure. A motor operated choke valve controls the pressure of the gas at the well Inlet. The wellhead annulus pressure (pressure downstream of the gas lift choke valve) varies from 1000 psig to 2200 psig max.

Field Background and Challenges

The gathering and production facilities include Remote Degassing Stations (RDS) and Central Degassing Station (CDS). The purpose of surface facilities at remote stations is to gather the crude from the wells and transport the crude to the nearest existing processing facility through transfer lines. Multiphase Pumps (MPP) are installed at Gathering stations to transport multiphase fluid through separate transfer lines. The proposed scheme for remote field operations consists of an Oil Gathering System to gather oil from multiple number of production wells to Multiphase Pumping (MPP) station through multi-port selector manifold (MSM). Multi-Phase Flow Meter (MPFM) will be provided for the well testing. The Multiphase Pump boosts the pressure of the production from wells to a level sufficient to deliver to the transfer line, approximately 21 Km away.

The well fluids form the well are gathered and collected in three remote degassing station RDSs. From the RDS the fluids are transferred through pig gable transfer lines to Central degassing stations for further processing.

The remote stations are manned during the daylight hours and are unmanned during night hours. The remote stations are operated using a DCS and can be operated from CDS control room after daylight hours using the "Remote" option using SCADA.

Wells are categorized as single completion, dual completion or artificial lift.

The reservoir sustainable production rate is maintained by using water injection (water injector wells) and gas injection (gas injector wells).

All wells are operated in accordance with the Well Integrity and Operations Standard. All well operations are planned and performed in compliance with all the applicable legislation and regulations. These may include but are not limited to directives/policy on HSE Performances, Ethical Conduct, Employees, Control and Finance.

Wells are further classified as oil producers, water supply, water injection, gas injection, aquifer reservoir observation, and disposal.

Wells Structure

Wells are equipped with surface safety valve (SSV) and Subsurface safety valve (SSSV). The Xmas tree is part of the wellhead and comprises a series of valves to allow the outflow of well fluids, and any auxiliary fluids to be controlled. Each well flow rate is regulated by a choke valve. Most choke valves are manually adjustable while other valves are automated for flow changes to be performed from the Local Control Panel / Control Room.

Each well is equipped with a wellhead at the surface level. The wellhead consists of three main parts; Casing Head Housing (CHH), Tubing Head Spool (THS) and Xmas Tree. The CHH and THS each contain valves that allow access to the annulus. The Xmas Tree comprises a series of valves to allow the outflow of well fluids, and any auxiliary fluids to be controlled. The well flow rate is regulated by a choke valve.

Choke valves

Choke valves are either positive or adjustable / motorized. The adjustable chokes can be either manual or automatic which allows the choke size change to be performed from the local control panel, RTU or the control room.

Figure 1—Typical Positive Choke Valve

Figure 2—Typical Adjustable Choke Valve

Figure 3—Motorized Choke Valve for oil and gas sides

Positive type chokes incorporate a fixed bean (similar to an orifice plate) that is available in a range of sizes graduated in sixty-fourths of an inch.

If there is a requirement to change the flow rate of a particular well it is necessary to remove the existing bean and replace it with one of a different size. Manual intervention is required.

As the bean in a choke can be abraded by solids in the well fluids (which would increase the size of the bean) there is a requirement to monitor bean performance as related to well flowrate and replace the bean from time to time.

Adjustable choke valves are installed in order to provide the means of flow regulation from the wells.

Most wells are completed with four casing strings.

The surface casing of 18 5/8 inch or 20 inch diameter, called conductor, is usually set to a below surface depth lower than 300 ft.

Set inside the surface casing is the 13 3/8 inch diameter casing to an approximate depth of 5000 ft, and a further 9 5/8 inch casing to a depth of about 8000 ft.

A 7 inch or 4 1/2 inch diameter liner hangs inside the 9 5/8 inch casing 100 ft from the bottom and set across the pay zone.

The space between two sets of casing is called annulus.

The purpose of the surface and intermediate casing strings is to seal the well, preventing any oil or gas leakage through the ground to the surface or into the aquifers.

Annulus A-B-C are present.

Annulus A: the annulus between the tubing and production casing.

Annulus B: the annulus between the production and the intermediate casing.

Annulus C: the annulus between the intermediate casing and surface casing.

Wells are categorized as:

Single completion (single tubing string);

Dual completion (two tubing strings);

Artificial lift using electrical submersible pumps or gas lift.

Surface arrangements for each well are generally similar.

Artificial lift Wells

Artificial Lift can be carried out using Electric Submersible Pump (ESP) or lift gas.

The Artificial Lift producers using ESP are single string and equipped with a down hole Electric Submersible Pump (ESP) to boost the flow from the original free flowing wells.

The principle of gas lift is that gas injected into the tubing reduces the density of the fluids in the tubing, and the bubbles have a "scrubbing" action on the liquids. The single completion gas lift as single completion well have one tubing for the oil extraction from the reservoir for the oil sending to the gathering manifold and one string for gas injection.

Wellheads

The wellhead is located on the top of the well casing strings and provides a seal for the annuli created by the different sized casings within the well, all of which terminate at approximately ground level.

On the casing head housing and tubing head spool of the wellhead various valves are located.

Dedicate valves provide access to the annuli created by the casing strings. Access is required to test the pressure build up in the annuli, to take fluid samples from the annuli and, when necessary, to bleed off pressure. Valves are often installed in tandem to provide positive shut-off. The termination point is either blanked or may be fitted with a needle valve and plugged off.

The Xmas tree incorporates five main valves and a wellhead cap for each string of the well. The valves are always gate-type valves.

The lowest valve is called the Lower Master Valve which, to avoid erosion, is only used in emergency situations, serves to isolate well fluids from the Xmas tree and must never be used for routine isolation of a well.

Above the lower master valve is the Upper Master Valve.

On the production tubing, above the Upper Master Valve, one or two Wing Valves are provided. One is the start of the well flow line.

On the top of the production tubing is the Swab Valve. This is used for wireline operations to provide isolation between the wireline equipment (the lubricator) and the well. The Bottom Hole Test Adaptor (BHTA) cap is removed (with the swab valve closed) and the wireline lubricator is attached in place of the cap. The lubricator provides a means of ingress and egress to the well tubing strings while the well remains under pressure.

Wellhead Control Panel

All producer wells have wellhead control panels. The SC-SSSV and SC-SSV close due to:

- Low pressure on actuation of low pressure pilot;
- High pressure on actuation of high pressure pilot;
- Detection of fire near the wellhead by melting the fire fusible plugs: one high pressure fire fusible plug installed in the hydraulic line to the SC-SSSV and four low pressure fusible plugs.

The instrumentation associated with the wellheads and flow lines includes local pressure and temperature gauges.

General Operating philosophy and requirements

Opening of Valves

Opening of the valves is a manual operation. Under no circumstances should any of the valves open automatically. The operating sequence is designed such that the SC-SSSV always opens before the SSV(s).

The SSV(s) can be manually opened and closed (locally and remotely) as long as the SC-SSSV is opened. All the actuators are spring loaded to close type (FTC; fail to close).

Control system design logic ensures that the above opening sequence is always maintained. Change in the opening sequence is not permitted.

The panel incorporates start-up bypass requirement to bypass low pressure signal. The bypass system is auto reset when the low pressure signal has been cleared. High/Low pressure status indicators are provided at the panel front.

The Wellhead Control Panel is used to control Surface Safety Valve(s) and Surface Controlled Sub-surface Safety Valve by applying pressurized hydraulic oil (pressure to open) to the valves' actuators. The SSV(s) and SSSV are fail closed valves and are normally kept in the "Open" position by energized low powered directional solenoids. De-energizing the solenoids will redirect the pressurized oil from the valves' actuators. The oil pressure shall be generated by an electric motor driven pump. Motor power rating shall be sufficient to generate the required pressure for opening the above valves.

The wellhead control system is totally independent of the flow line fluid using clean hydraulic oil in a closed, leak tight circuit. This is required to eliminate the risk of contamination of the actuator and control circuit by well fluid and to ensure reliability and durability of WHCP function.

The SSV(s) & the SC-SSSV usually operate at different pressures. Two pressure ranges are provided for hydraulic supply, one pressure range for SSV(s) & second pressure range for SC-SSSV.

The wellhead control system allows manual operation of SC-SSSV and SSV(s). The controls are designed for fail safe operations and easy to service without removing the hydraulic relays out of the system for maintenance and calibration. System is designed both for user and environment friendliness.

The control panel is solar powered designed as a stand-alone wellhead control system both for the operation of single or dual completion oil producers as given in the material requisition.

Control system design logic ensures that the above opening sequence is always maintained.

The SSSV mounted on the tubing of the wellhead, is hydraulically actuated and is open during normal operation.

A high pressure fusible plug mounted on the hydraulic actuation line allows, when it melts (94°C) in case of fire, the SSSV closing, while around the wellhead four low pressure fusible plugs are mounted by separate headers.

SSSV closes after a time delay and only when the ESD is activated (the ESD push button mounted in front of the panel, remote ESD actuation or fusible plugs actuation).

A remote ESD point is installed at least 30 metres downwind from the wellhead at a suitable location & at well fence gate.

Solar System/Passive Shelter

The power supply for the panel is 24 VDC, -10%, +15%, negative earthed. This power supply is fed by the SOLAR power system supplied by others. The panel receives power by the non-redundant separate feeder for hydraulic pumps and control panel.

The solar system is provided at the wellhead station to meet the power requirement for the operation of the wellhead control panel, PLC and SCADA panels and limited lighting. An isolated, thermally insulated enclosure is provided as a passive shelter for housing the PLC, SCADA panels. The solar system and the passive shelter with the thermo-siphon based water cooling arrangement, which maintains the temperature inside the shelter at the required operating level, day and night, leave the wellhead operation independent of external utilities.

High and low fusible plugs are hydraulically powered by the wellhead panel hydraulic system. Whenever a fire occurs around the well, the hydraulic pressure loop of the LP fusible plugs or the hydraulic pressure loop of the HP fusible plug gets depressurised causing the shutdown of the wellhead.

In addition to the above safety protections, the wellhead is also provided with an Emergency Shutdown switch. A remote ESD point is installed at least at 30 metres downwind from the wellhead at a suitable location & at well fence gate.

Single Completion Gas Lift Well

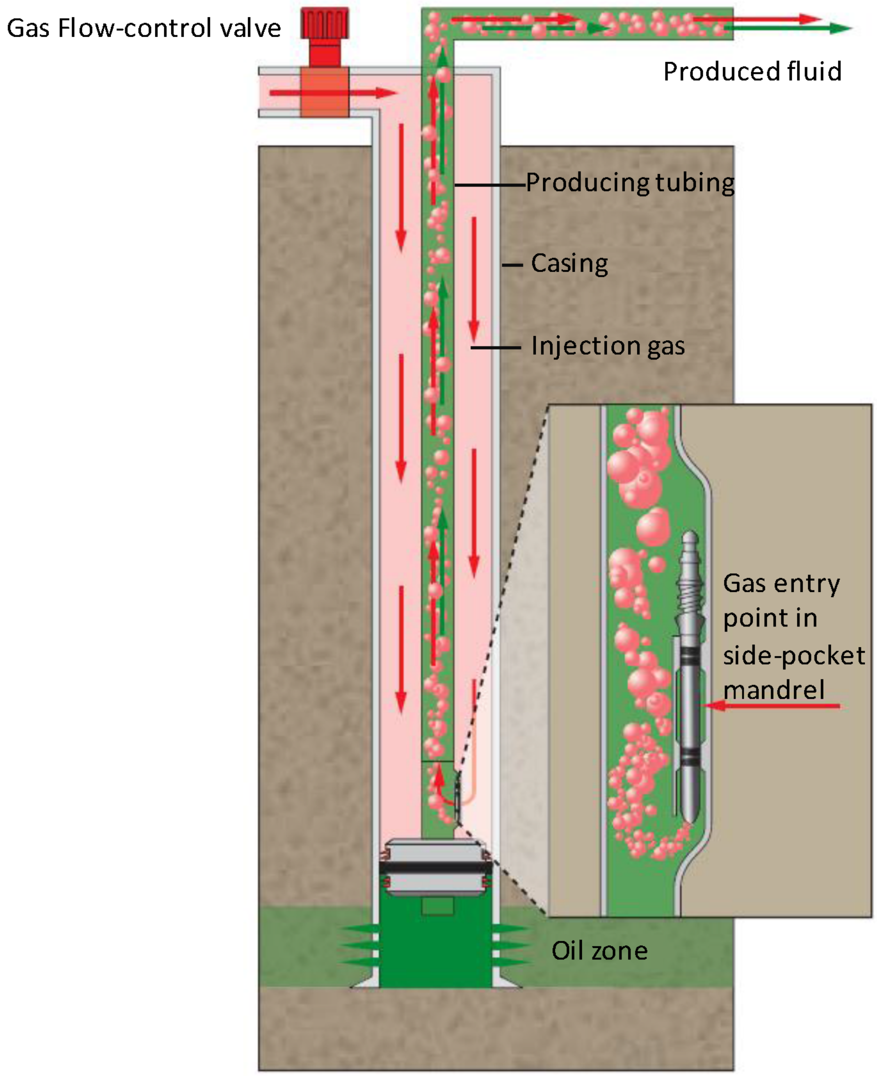

Figure 4—Typical Single Oil production Gas lift Completion Type

Introduction

The principle of gas lift is that gas injected into the tubing reduces the density of the fluids in the tubing, and the bubbles have a "scrubbing" action on the liquids. The single completion gas lift as single completion well have one tubing for the oil extraction from the reservoir for the oil sending to the gathering manifold and one string for gas injection. Unlike the single completion, in the single completion gas lift, oil extraction is carried out by injecting compressed lean gas by the means of a second line (gas lift piping).

Lean gas comes from the nearby plant to ensure uninterrupted supply of lift gas for the wells.

The lean gas from the compressor discharges is routed to the pressure let-down station for gas lift where pressure is reduced from 400 barg to 180 barg.

The gas is then routed to the wells through a pipeline network including a 6″ main header and 4″ sub-header with the 3″ tap-offs to the individual wells.

Surface arrangement

Gas Side

The wellhead 2″ piping with double block valves is located at about 100m away from the wellhead, inside a fixed fenced area, for the well isolation during maintenance activities. Also a shutdown valve (SDV) is provided in order to isolate the pipeline from the wellhead in the event of fire due to the oil spill in the well area and ruptures in the piping inside the wellhead fenced. A Y strainer, a vortex flow meter and a pressure control valve are also provided on the gas side.

The gas flow to the well is regulated at the pressure required for the gas lift. This is achieved by pressure controller and a motorized choke valve.

Double block valves are provided for isolation at the wellheads at the Xmas tree end and at the gas lift piping battery limit end.

A check valve is provided between the manual isolation valve and the gas side wing valve.

The surface safety valve hydraulically actuated, located between the check valve and the wing valve (side gas) is handled from the wellhead control panel.

2″ manual depressurising provision for the wellhead gas lift piping is given at the downstream of the SDV, downstream of the remote operated choke valve and downstream of the check valve discharging to the vent boom.

The vent boom is provided with a flame arrestor. It should be noted that this vent is for depressurising the well head piping alone and not for gas flow pipelines and flow line from the gas injection compressor at NGIP, which if required needs to be done inside NGIP limits.

The provision for nitrogen purging of the injection line is also given.

A DPI indicator measures the pressure drop across the Y strainer. The flow meter indication is pressure and temperature compensated. The remote operated choke opening is adjusted based on this flow indication.

Oil Side

The 3″ string producer is equipped with wing valve, surface safety valve (SSV) (hydraulically actuated), choke valve for the flow regulation, pressure transmitter and related local pressure indicator (PI), pressure transmitters (PT), check valve (wafer type) to prevent oil back flow from the gathering manifold in case of the well shutdown. The sub surface safety valve (SSSV) installed on the tubing is open in normal operation. Opening/closing (this valve is hydraulically actuated) is handled from the wellhead control panel LCP.

Wellhead control panel LCP

The PLC based wellhead control panel is envisaged with solar power. The wellhead control panel functions are the following:

- To control the opening and closing of the gas lift SDV at the battery limit (SDV), the surface safety
- valves on gas side (SSV) and oil side (SSV).
- To open subsurface safety valve SSSV.
- To take care of the well safety requirements like fire, gas and ESD activation.
- To provide interface through the data link to the remote NGIP control room for monitoring the well parameters as well as facilitate remote operation of the well.

RTU/PLC XCP

Wellheads have also provision for remote operation of the following actions through the data link between DCS at NGIP and RTU/PLC at the wellheads:

Adjust gas lift choke (gas side choke valve) by the remote set point for gas lift pressure Well ESD command

Well testing facilities

Production from the inlet system at CDS is diverted into five Separator Trains, in parallel configuration. Crude oil from 2nd Stage Production Separators is routed by a common distribution header to three vertical gas/liquid separators named Degassing Boots.

The production from each local oil well can be routed to one of the new three phases test separators and one MPFM located at CDS plant.

The purpose of a Test Separator is to allow the component flows (oil, gas and water), from a single well or source, to be measured and sampled. Any changes in a well's test results are plotted over a given time to give an indication of the change in reservoir conditions in the area from which the well is flowing.

Test separators are horizontal pressure vessels equipped with an internal weir, which allows settling of water and sludge in the first section of the vessel. Oil overflows the weir and collects in the oil compartment.

Each test separator is provided with two control and metering systems (one for low range and one for high range) in parallel installation for either gas or liquid outlet flow rates. The operating pressure is the same as the 1st Stage Production Separator (i.e., 17.2 barg).

The operator manually selects from the control room the proper control and metering systems according to the expected ranges of liquid and gas flow from the well under testing.

Testing Facilities

MPFM

The Multiphase Flow Meter (MPFM) is a system to measure a flow rate of multiphase flow (oil, gas and water) and is used for testing well fluids from the Inlet Manifold.

The two installed Multiphase Flow Meters FQI (MPFMs) allow the testing of two wells at a time during normal operation.

Each MPFM is always kept online, even when the testing of wells is not required.

The oil, gas and water flow rates, pressure and temperature of the well tested are transmitted to the BQ Control Room. Local on-site indication is also provided.

After metering the fluid is sent to the inlet of the MPP.

The MPFM is provided to measure the individual oil, water and gas flow rates of three phase crude. The diversion of the well under test to the MPFM is carried out manually. In order to avoid stagnant flow conditions to the header and MPFM, one well is always aligned through the MPFM always.

Temperature and pressure compensation are provided for the gas flow rate. The flow rate and totalized flow indication are available on the control system workstation in CCR.

MPFM draining is performed through the drain line to the drain header.

Double block-and bleed 6″ NPS connections are fitted to allow a portable test separator to be connected in order to verify the MPFM performance.

Meter Arrangement

The Multiphase Flow Metering System is designed to measure the individual flow rates of oil, water and gas (temperature and pressure compensation are provided for the gas flow rate); mass and volume calculations, cumulative measurements and GVF, GLR, GOR and water cut calculations.

The System consists of the following main components:

- Field Sensor Unit;
- Field Control Unit;
- Human-Machine lnterface (HMI).
- Calculation is performed in the MPFM HMI; the final data are transferred to the DCS for data logging.
- Double block-and-bleed 6″ NPS Portable meter prover connections are available for proving the MPFM if/as required.
- Control Description
- The diversion of the well being tested to the MPFM through one Test Header is performed remotely from Control Room; the operation can be also carried out locally in the field.
- An alarm is generated when the operating point falls beyond the meter envelope.

Field Sensor Unit

The Field Sensor Unit is installed on the MPFM inlet line and measures the parameters used in the flow calculations of oil, gas and water.

The field sensors are suitable for all the following flow regimes:

- Stratified flow;
- Bubbly flow;
- Churn flow;
- Slug flow.

The MPFM is capable of functioning and operating in the stream condition of 0-100% Water in Liquid Ratio (WLR) and 0-99% Gas Void Fraction (GVF) and all flow regimes, unaffected by paraffin or asphaltene deposition.

The in-line MPFM can perform self-verification checks (e.g. continuous system status self-checks, pressure measurement hardware, dynamic self-verification, etc.)

The overall accuracy of MPFM is as follows:

- Gas flow rate ±5% to ±10% relative
- Oil flow rate ±5% to ±10% relative
- Water flow rate ±3% to ±8% absolute
- Total fluid flow rate ±2% of reading
- Repeatability ±5% relative for each of the phases ±5% relative for WLR

Objective

Activation of the inactive wells looking into out of the box solution. The cost effective solution was required for timely reactivation of shut-in wells using in house resources. A Bullheading line was readily available solution for the execution.

Challenges/ Driver

- Huge cost for well intervention
- Commuting to the site.
- Extra manpower and resources requirements
- The commissioning of gas lift wells presents challenges in unloading the liquid column to allow lift gas from the annulus into the tubing. Occasionally, this process is unsuccessful, leading to back pressure on the gas side and preventing lift gas flow into the well.

Solution

- To address this issue, a 1-inch piping connection was established between the gas side and the oil side. Utilizing the available piping for rocking, rather than relying on conventional external sources, resulted in significant time and cost savings as well as timely activation of the well to enhance production. The wells have been successfully rocked.

Well Rocking

Bull heading a gas lift well is a well intervention technique used to force fluids down the tubing or annulus to clear blockages, remove hydrates, or displace unwanted fluids

Well rocking can be recommended when:

- Injection pressure at valve depth is lower than tubing pressure. So it is possible that first valve is too deep to be reached under the current operational conditions and hence unloading operations cease before reaching the operating depth. Rock the well is one of the temporary options.
- As part of well troubleshooting to remove temporary GL valve blockages instead of retrieving gas lift valves which save wire line operation and eliminate well intervention risks.
- Sometimes during early commissioning, gas injection is directed to tubing side purging the gas line from any sand or debris instead of directing these impurities to the gas line which could damage or plug gas lift valves prior to commissioning.

Purpose of Bull Heading in Gas Lift Wells

1. **Remove Blockages or Hydrates**
   - Gas lift wells can develop hydrate plugs or wax/paraffin deposits, especially in colder environments or during shut-ins.
   - Bull heading high-pressure fluid (usually treated water, diesel, or methanol) helps dislodge and push these blockages back into the formation or to the surface.
2. **Unload Kill Fluid or Heavy Liquids**
   - After a well kill or workover, heavy fluids may remain in the tubing.

- Bull heading helps displace these fluids to allow the well to resume production or gas lift operation.

3. **Stimulate Flow or Restart a Dead Well**
   - If a gas lift well has stopped flowing due to liquid loading or insufficient lift gas, bull heading can re-establish flow by clearing the tubing and re-energizing the wellbore.
4. **Clean Out the Tubing or Annulus**
   - Used as a preventive maintenance or troubleshooting step to clean out debris, scale, or sand that may be affecting gas lift valve performance.

Execution Strategy

Preliminary Checks (Oil Side)

- Confirm wellhead RTU is in service
- Confirm all instruments upstream and downstream of choke valve are in service for oil line
- Check hydraulic oil tank level and refill if necessary
- Confirm that the Hydraulic Pump is available
- Confirm that all bleed, vent and drain on surface equipment are closed
- Confirm that all bleed, vent and drain flow lines are closed
- Confirm that Subsurface Safety Valve and Surface Safety Valves are open
- Confirm that the choke valve is open.
- Confirm that all instrument isolation valves are opened with instruments in service

Preliminary Checks (Gas Side)

- Confirm open: SDV on lift gas manifold upstream of FCVgas choke valve
- Confirm locked open: · 2″ ball valves downstream of SDV on lift gas manifold and downstream of FCV gas choke valve
- Confirm closed all vent and drain valves of gas system

Procedural Steps

- Confirm Utilities and upstream (Gas line) lined up to wellhead.
- Confirm Wellhead and its auxiliaries are working.
- Close Wing Valve, Open Isolation Valve between gas and oil sides.
- Inject gas down the tubing until tubing head pressure is equal to gas lift manifold pressure or to the value advised by Gas Lift Team.
- Bleed the tubing pressure off very quickly while maintaining gas injection into the casing annulus.
- Apply gas lift on the tubing (TBG) side using the rocking line until the TBG pressure reaches 1300-1400 psig. The objective is to reduce the hydraulic head against the valve depth.
- Minimize flow line pressure before proceeding with a well-flowing trial.

- Increase casing pressure to the maximum while keeping the TBG side closed.
- Proceed with opening the wing valve to flow the well while maintaining the production choke at 32/64″ & GL injection rate (0.3 – 0.4 MMSCFD).
- The well starts to flow.

Restriction

This procedure should be attempted if the well has no sand problems and no damage is inflicted to the completion or to the surface facilities.

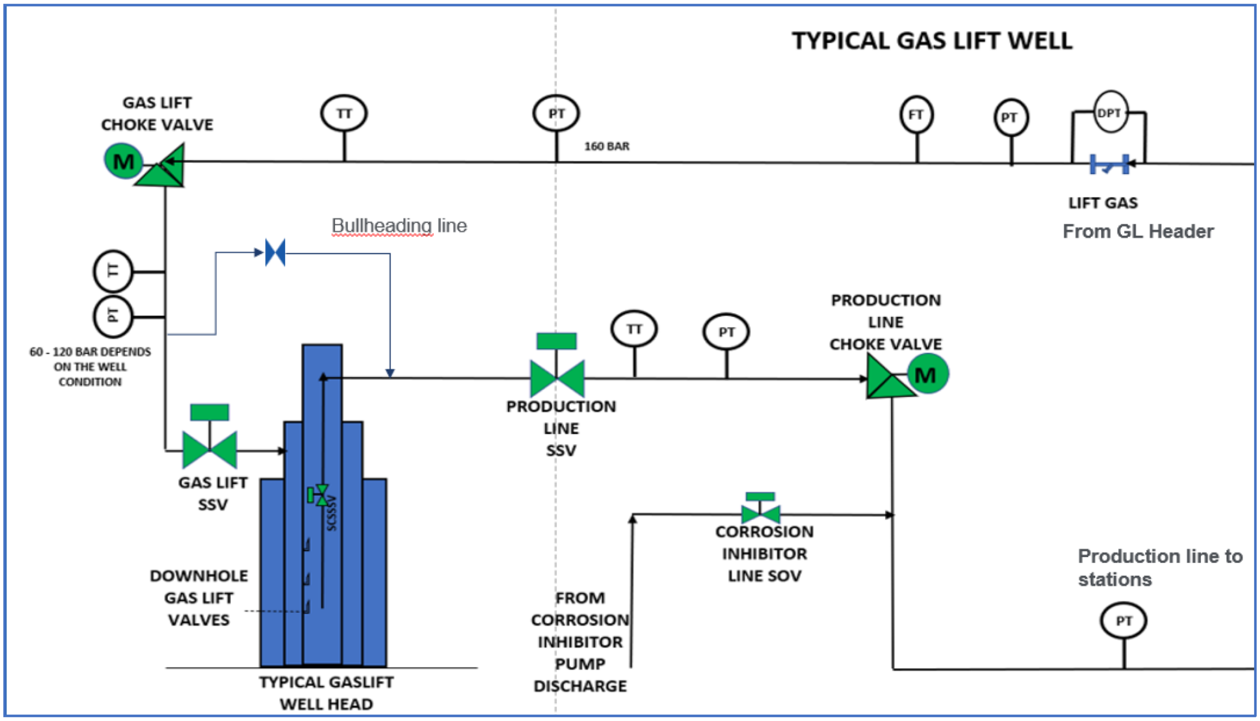

Figure 5—Typical Gas Lift Well

General Benefits

Lessons learned highlighted the benefits of minimizing external resources and reducing time consumption in resuming production. The innovative solution of utilizing the bullheading/rocking line proved to be cost-effective. The benefits achieved from this exercise, which will continue to be advantageous in the future, include:

- Reduction of greenhouse gas emissions (3 Metric Tons CO2 equivalent)
- Avoidance of intervention costs amounting to 1 million USD
- Production gain of 10,000 barrels of oil, preventing production loss from gas lift wells
- Well availability exceeding 98%
- Maintenance of inactive strings KPI below 2%
- Ensuring well integrity
- Achievement of project milestones on time
- Avoidance of costs and time incurred due to prolonged downtime.

Conclusion

The implemented solution is highly effective in ensuring continuous production, reducing GHG emissions leading to NET zero mindset and can be replicated to other existing facilities in the company and worldwide. Adding simple piping at wellhead resulted in the way forward activating the wells without using the external resources.

This implementation will enhance the availability of wells in short time span and proved helpful in contributing under the pillar of Sustainability – Reduction in flaring/GHG emissions.

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Glossary

MPP – Multiphase pump
MPFM – Multiphase fluid metering
BOPD – Barrels Oil per day
BWPD – Barrels water per day
WHCIP – Well head closed in pressure
TSS – Total suspended solids
PTS – Potable Test Separator
MSM – Multi Selector Manifold
KOD – Knock out drum
FDIS – Field Data Integration System
RCA – Root Cause Analysis
DCS – Distributed Control System
RDS – Remote Degassing Station
CDS – Central Degassing Station
F&G – Fire & Gas
VMS – Visual Management Board
PLC – Programmable Logic Controller
KOD – Knockout Drum
DGS – Downgraded Situation
RTU – Remote Telemetry Units
ACM – Advanced Communication modules
ICS – Instrumented control system
CHH – Casing Head Housing
THS – Tubing Head Spool

References

Gas lift wells Operating Manual, DCS, RTU and PI system data